

❁ APPENDIX D ❁

Calendar Math

Dictionaries define the calendar almost as if it were a machine: "a system for fixing the beginning, length, and divisions of the civil year." But in every society calendars are much more than that. People experience time as both linear and circular. On the one hand, it marches remorselessly from birth to death, a vector with fixed endpoints and a constant velocity. On the other hand, time is cyclical, with the wheel of the seasons endlessly spinning, and no clear end or beginning. Calendars are records of a culture's attempt to weight and reconcile these different visions.

In early European societies, the end of the year was regarded as dangerous: a period when the calendar literally runs out of days, the landscape is blanketed by night and cold, and nobody can be truly certain that the heavens would usher in a new year. Embodying that mysterious time when the end of the calendar somehow looped round and rejoined itself at the beginning, Romans celebrated Saturnalia, an upside-down week when masters served their servants and slaves held the great offices of state. The Christian calendar bracketed the strange, perilous final days of the year on one end with the birth of Christ, symbol of renewal, on December 25, and on the other with Epiphany, the day when the three kings recognized the infant Jesus as the Savior, another symbol of renewal, on January 6. Christmas and Epiphany bridge the dangerous gap between the end of one year and the beginning of the next.

The Mesoamerican calendar also tied together linear and cyclical time, but more elaborately. In its most fully developed form, at the height of Maya power, it consisted of three separate but interrelated calendars: a sacred tally known as the *tzolk'in*; the *haab*, a secular calendar based, like the Western calendar, on the rotation of the sun; and the Long Count, a system that, among other things, linked the other two.

The sacred calendar is both the calendar most dissimilar to Western calendars and the most important culturally. Each day in the *tzolk'in* had a name and a number, in somewhat the same way that one might refer to, say,

"Wednesday the 15th." In the Western calendar, the day names (e.g., Wednesday) run through cycles of seven, making weeks, and the day numbers (e.g., the 15th) run through cycles of 28, 30, or 31, making months. The *tzolk'in* used the same principle, but with less variation in the lengths of the cycles; it had a twenty-day "week" of named days and a thirteen-day "month" of numbered days. The analogy I am drawing is imprecise; what I am describing as the *tzolk'in* "week" was longer than the "month." But just as Thursday the 16th follows Wednesday the 15th in the Christian calendar, 10 Akbal would follow 9 Ik in the *tzolk'in*. (The Maya had a twenty-day "week" in part because their number system was base-20, instead of the base-10 in European societies.)

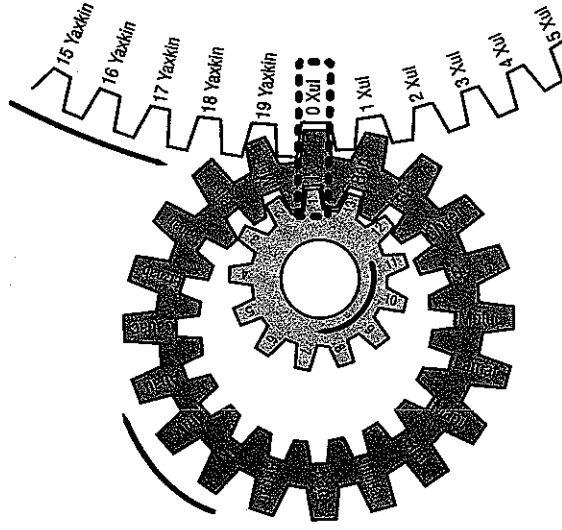
Because the *tzolk'in* was not intended to track the earth's orbit around the sun, its inventors didn't have to worry about fitting their "weeks" and "months" into the 365 days of the solar year. Instead they simply set the first day of the year to be the first day of the twenty-day "week" and the thirteen-day "month," and let the cycle spin. In the language of elementary school mathematics, the least common multiple (the smallest number that two numbers will divide into evenly) of 13 and 20 is 260. Hence, the *tzolk'in* had a length of 260 days.

In the Western calendar, a given combination of named and numbered days, such as Wednesday the 15th, will occur a few times in a calendar year. For instance, in 2006 the 15th of the month falls on Wednesday three times, in February, March, and November; in 2007 Wednesday the 15th occurs just once, in August. The irregular intervals are due to the differing lengths of the months, which throw off the cycle. In the *tzolk'in*, every "month" and every "week" are the same length. As a result, "Wednesday the 15th"—or 1 Imix, to give a real example—in the *tzolk'in* recurs at precise intervals; each is exactly 13×20 or 260 days apart.

Many researchers believe the movements of Venus, which Mesoamerican astronomers tracked carefully, originally inspired the *tzolk'in*. Venus is visible for about 263 consecutive days as the morning star, then goes behind the sun for 50 days, then reappears for another 263 days as the evening star. It was a powerful presence in the heavens, as I noted in Chapter 8, and a calendar based on its celestial trajectory would have shared some of that power. Within the sacred year, every day was thought to have particular characteristics, so much so that people were often named after their birth dates: 12 Eb, 2 Ik, and so on. In some places men and women apparently could not marry if they had the same name day. Days in the *tzolk'in* had import for larger occasions, too. Events from ceremonies to declarations of war were thought to be more likely to succeed if they occurred on a propitious day.

Because people also needed a civil calendar for mundane purposes like knowing when to sow and harvest, Mesoamerican societies had a second, sec-

ular calendar, the *haab*: eighteen "months," each of twenty days. (Unlike the *tzolk'in*, which counted off the days from 1, the *haab* months began with 0; nobody knows why the system was different.) Simple arithmetic shows that eighteen twenty-day months generates a 360-day year, five days short of the requisite 365 days. Indians knew it, too. Rather than sprinkling the extra five days throughout the year as we do, though, they tacked them onto the end in a special "month" of their own. These days were thought to be unlucky—it was as if the year ended with five straight days of Friday the 13th. Although the ancient Maya knew (unlike their contemporaries in Europe) that the solar year is actually $365\frac{1}{4}$ days, they did not bother to account for the extra quarter



The Mesoamerican calendar was both more complex and more accurate than the European calendars of the same period. It consisted of a 365-day secular calendar, the *haab* (right), much like contemporary European calendars. The *haab* was tied to the second, sacred calendar, the *tzolk'in* (left), which was unlike any Western calendar. With a "week" of twenty named days and a "month" of thirteen numbered days, the *tzolk'in* produced a 260-day "year." Mesoamerican societies used both simultaneously, so that every date was labeled with two names (1 Xul 0 Xul in the drawing). I have not rendered the *haab* as a wheel-within-wheel like the *tzolk'in*, even though it, too, had perfectly regular "weeks" and "months." This is because the *haab* had to fit the 365-day solar year, which forced Maya calendar designers to spoil their system by tacking on an irregular, extra-short month at the end.

day; there were no leap years in Mesoamerica. The failure to do so seems surprising, given that their astronomers' mania for precision had led them to measure the length of the lunar month to within about ten seconds.

With two calendars, every day thus had two names, a sacred *tzolk'in* name and a civil *haab* name. Usually the Maya referred to them by both at once: 1 Ix o Xul. The two different calendars, each perfectly regular (but one more regular than the other), marched in lockstep, forming what is now called the Calendar Round. After one 1 Ix o Xul, there would not be another 1 Ix o Xul for 18,980 days, about fifty-two years.

By describing dates with both calendars Mesoamerican societies were able to give every day in this fifty-two-year period a unique name. But they couldn't distinguish one fifty-two-year period from its predecessors and successors—as if the Christian calendar couldn't distinguish 1810, 1910, and 2010. To avoid confusion and acknowledge time's linear dimension, Mesoamerican societies invented the Long Count, which counts off the days from a starting point that is believed to have been in mid-August, 3114 B.C. Long Count dates consisted of the number of days, 20-day "months," 360-day "years," 7,200-day "decades," and 144,000-day "centuries" since the beginning. Archaeologists generally render these as a set of five numbers separated by dots. When Columbus landed, on Tuesday, October 11, 1492, the Maya would have marked the day as 11.13.12.4.3, with the "centuries" first and the days last. In the *tzolk'in* and *haab*, the day was 2 Akbal 6 Zotz.

Although extant Long Count dates have only five positions for numbers, the Maya knew that eventually that time would pass and they would have to add more positions. Indeed, their priestly mathematicians had calculated *nineteen* further positions, culminating in what is now called the *alautun*, a period of 23,040,000 days, which is about 63 million years. Probably the longest named interval of time in any calendar, the *alautun* is a testament to the grandiosity of Mesoamerican calendrics. Just as the *tzolk'in* is one of the most impeccably circular time cycles ever invented, the Long Count is among the most purely linear, an arrow pointing straight ahead for millions of years into the future.

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